

Anomaly Detection and Traffic Prediction with Isolation Forest (IF) and Long Short-Term Memory (LSTM) Models

Stefan-Darian CIRNAT
Coordinator: Virgil DOBROTA

E-mail: stefandarian@yahoo.com

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Description

A system for monitoring and analyzing network traffic using open-source tools. It collects network metrics from GNS3 using Prometheus, displays them in Grafana, and applies AI models for anomaly detection and traffic prediction.

Motivation

Existing network tools are often expensive and lack good documentation. This project was created as an open-source solution with detailed documentation and potential for future expansion.

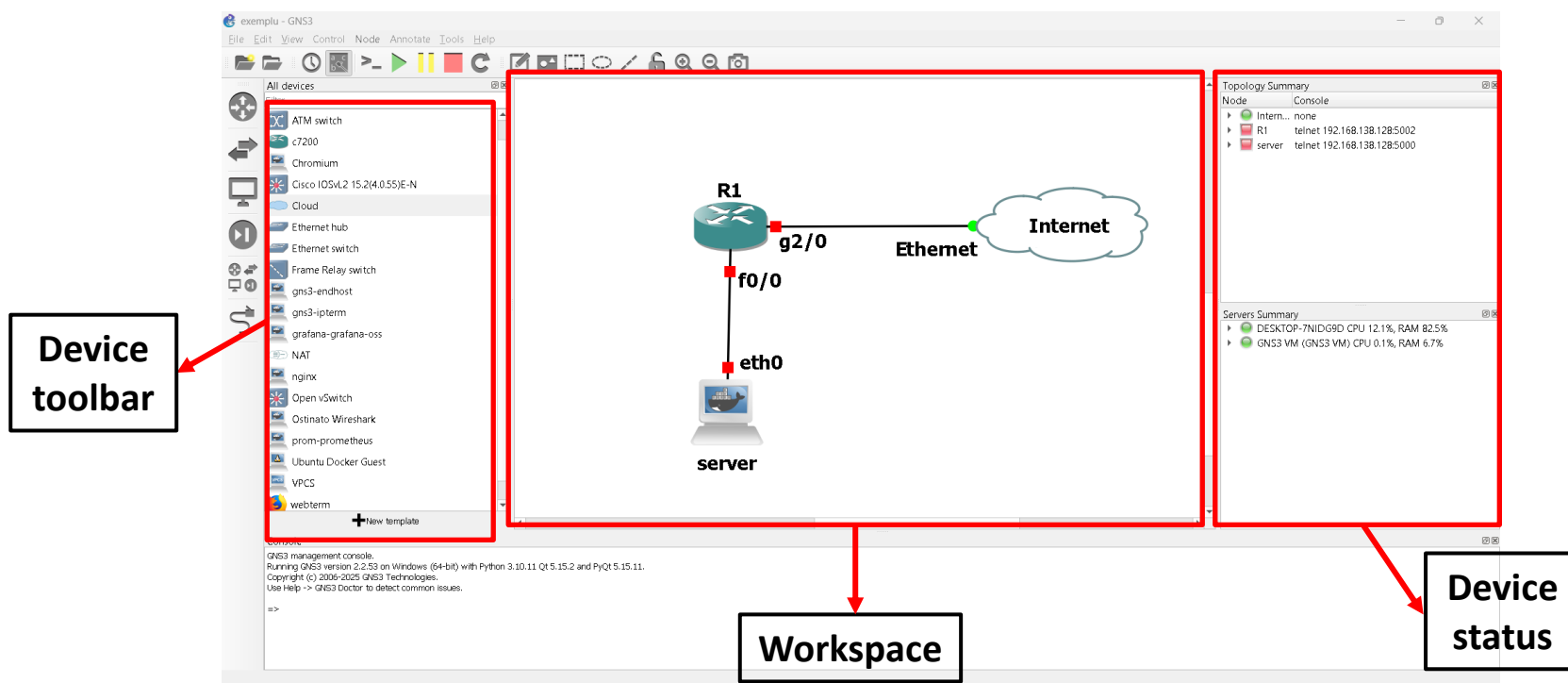
Purpose

Develop an open-source solution for network monitoring and intelligent traffic analysis using AI.

Main Objectives

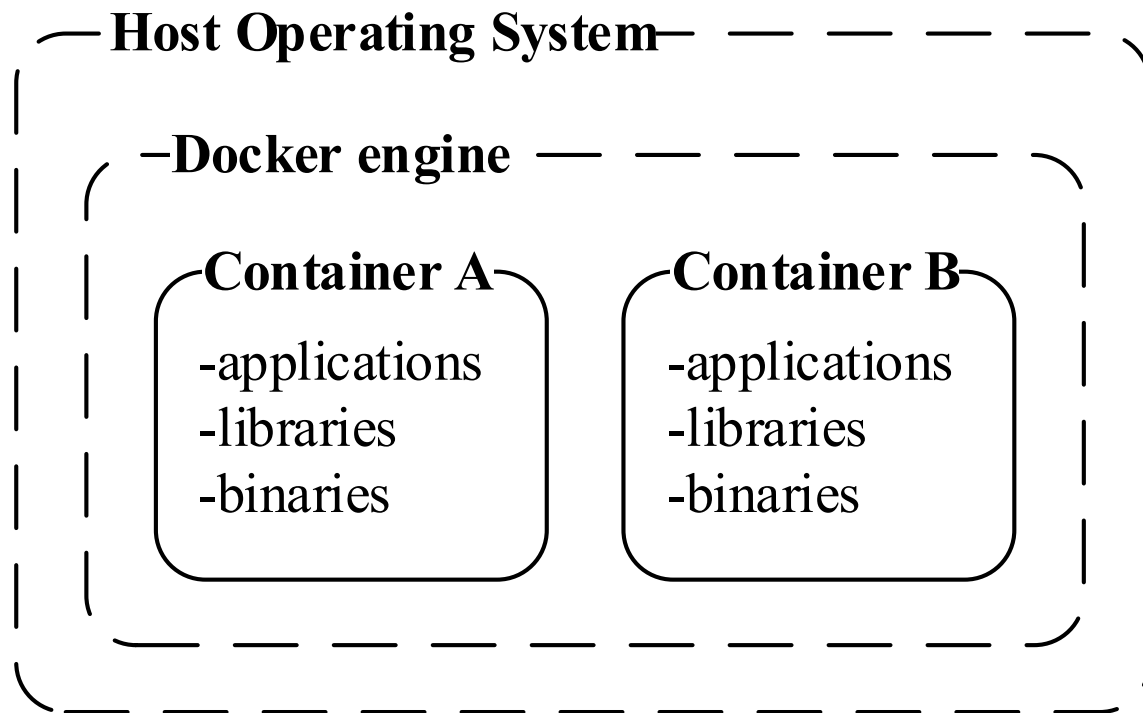
- Monitor traffic in real time
- Use open-source tools
- Detect anomalies and predict traffic using AI

- Realistic network emulation
- Integration with Docker containers
- Cost-effective and flexible
- The components run inside a Virtual Machine (ex. VMware) integrated into the GNS3 interface



Docker Containers

Docker container architecture

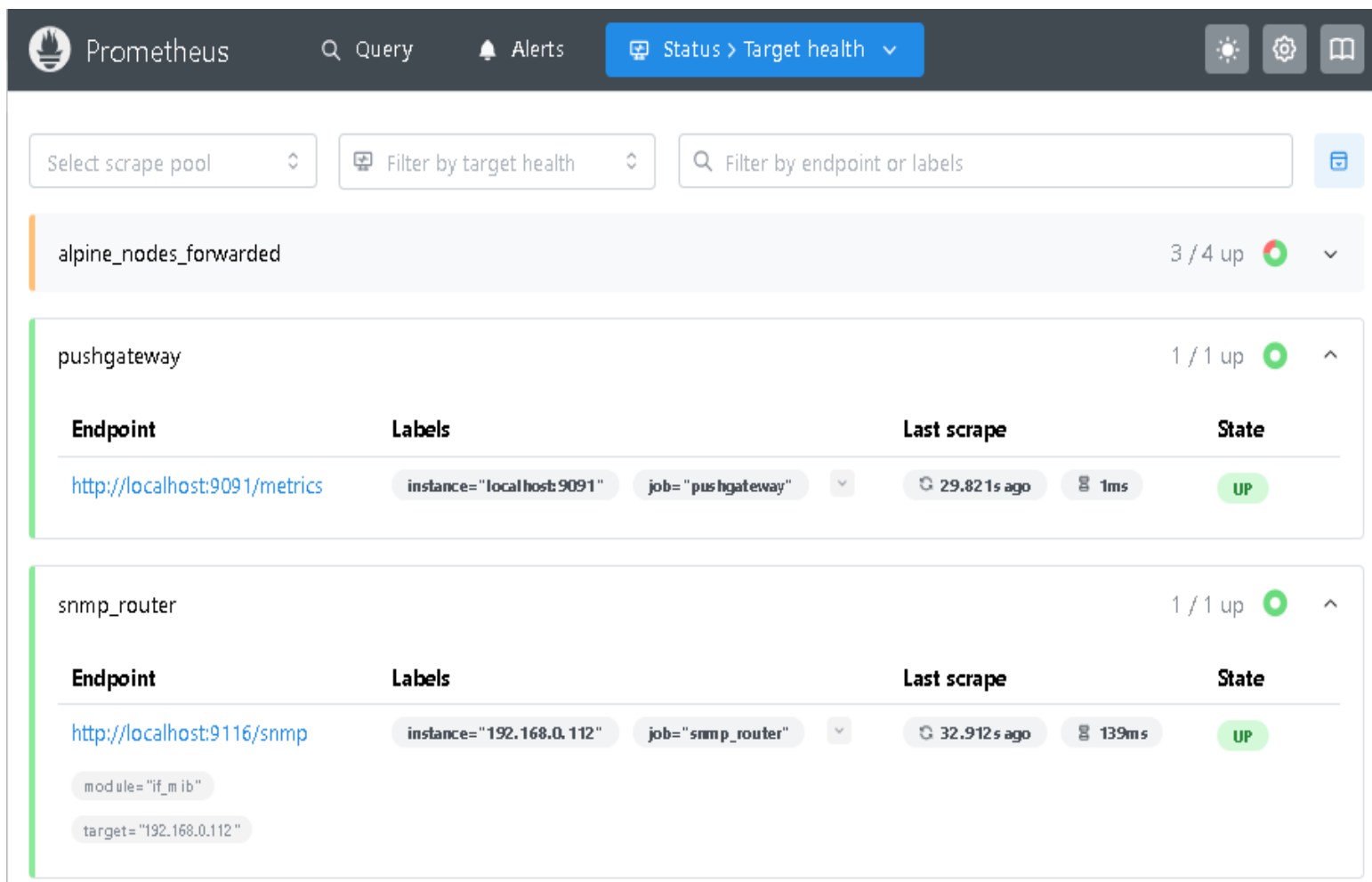


Features:

- portable on any type of OS
- Lightweight & fast (vs. VMs)
- GNS3 integration

Monitoring Tools: Prometheus, Grafana and Pushgateway

- Prometheus graphical interface



The screenshot displays the Prometheus web interface. At the top, there's a navigation bar with 'Prometheus', 'Query', 'Alerts', and 'Status > Target health'. Below this, there are filters for 'Select scrape pool', 'Filter by target health', and 'Filter by endpoint or labels'. The main content area shows two target groups: 'alpine_nodes_forwarded' (3/4 up) and 'pushgateway' (1/1 up). The 'pushgateway' group is expanded, showing a table of targets.

Endpoint	Labels	Last scrape	State
http://localhost:9091/metrics	instance="localhost:9091" job="pushgateway"	29.821s ago 1ms	UP

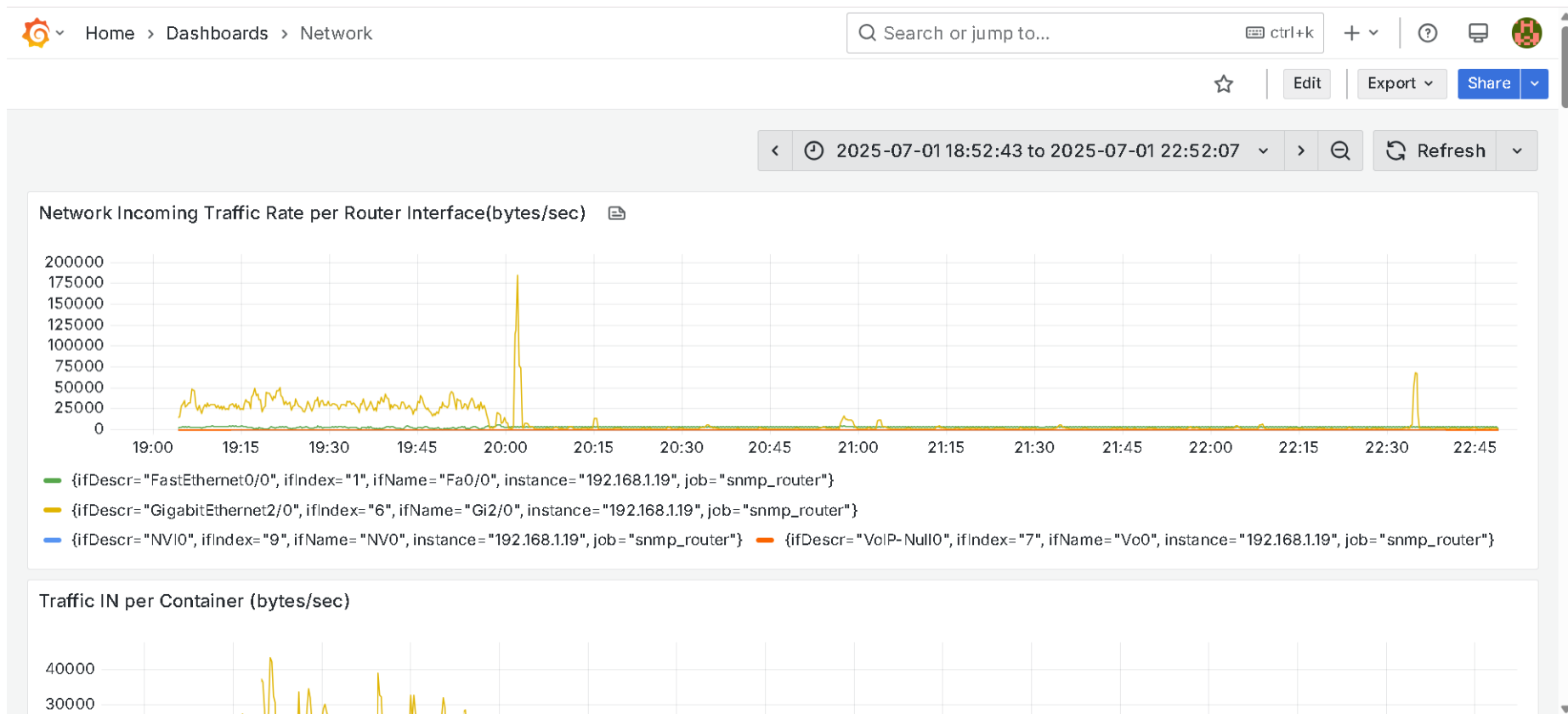
Below the 'pushgateway' group, there's another target group 'snmp_router' (1/1 up). It also shows a table of targets.

Endpoint	Labels	Last scrape	State
http://localhost:9116/snmp	instance="192.168.0.112" job="snmp_router"	32.912s ago 139ms	UP

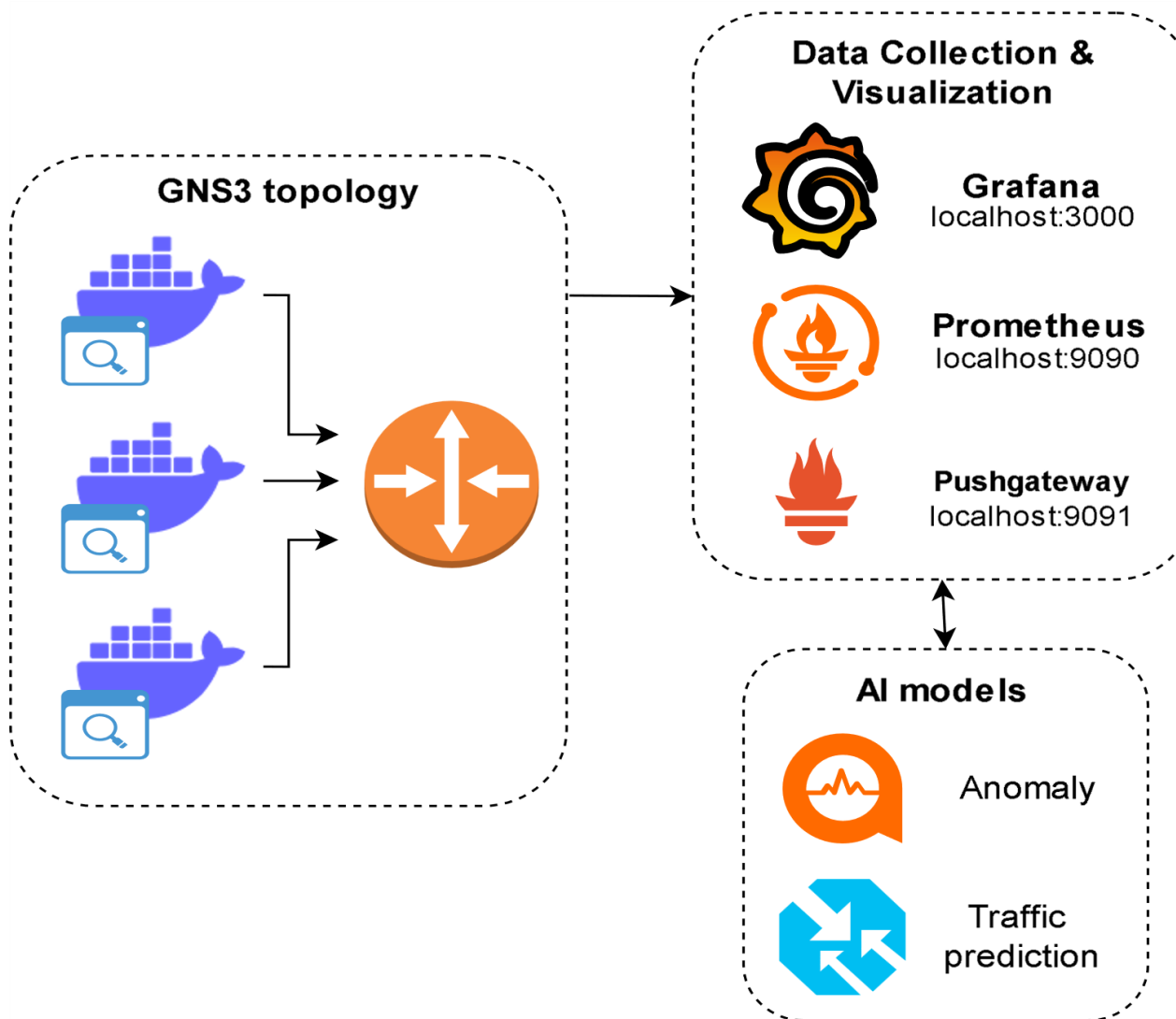
Additional labels for 'snmp_router' include 'module="if_mib"' and 'target="192.168.0.112"'.

Monitoring tools: Prometheus, Grafana and Pushgateway

- Grafana dashboard



System Architecture

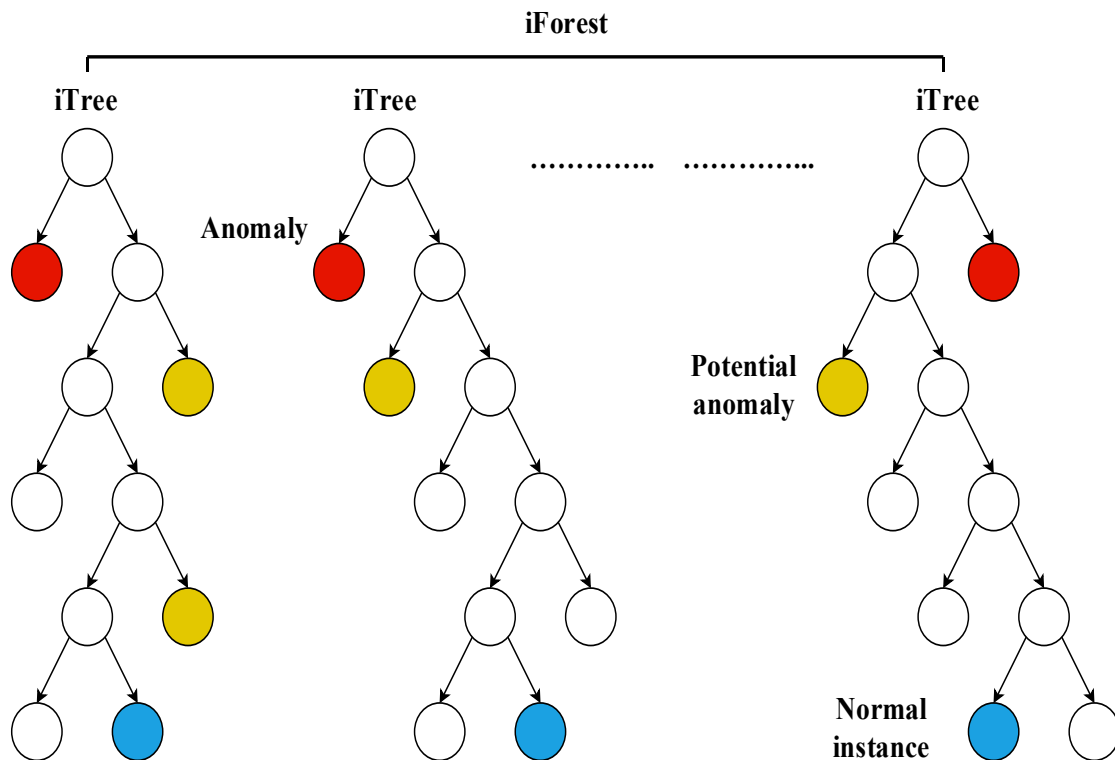


Isolation Forest (IF) for Anomaly Detection

- IF: an **unsupervised** ML algorithm, isolates anomalies, based on the assumption that anomalies are **few and different**.

- Algorithm steps:

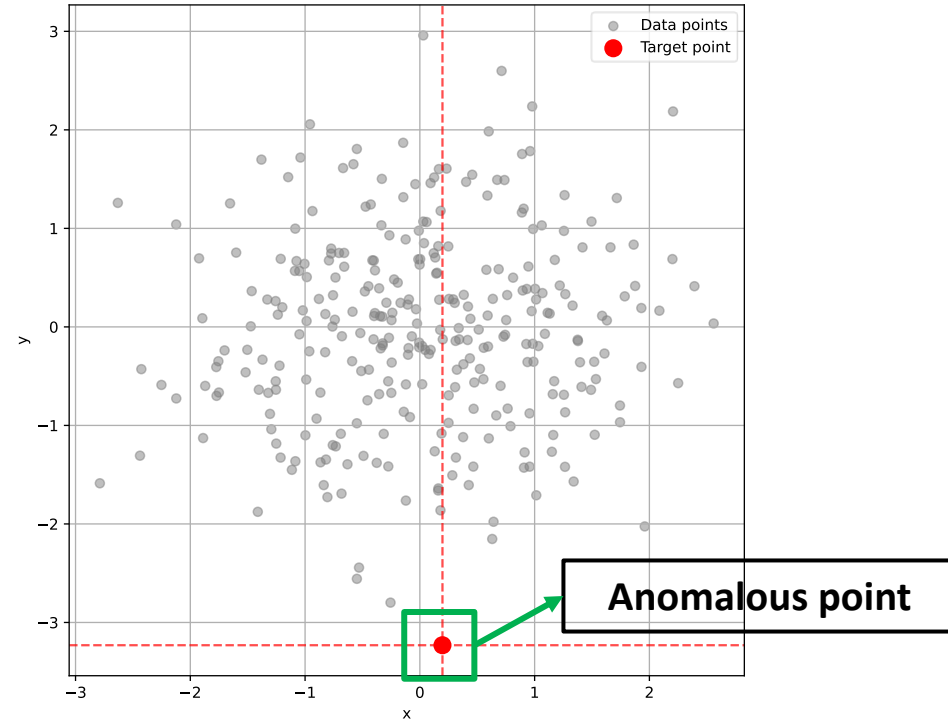
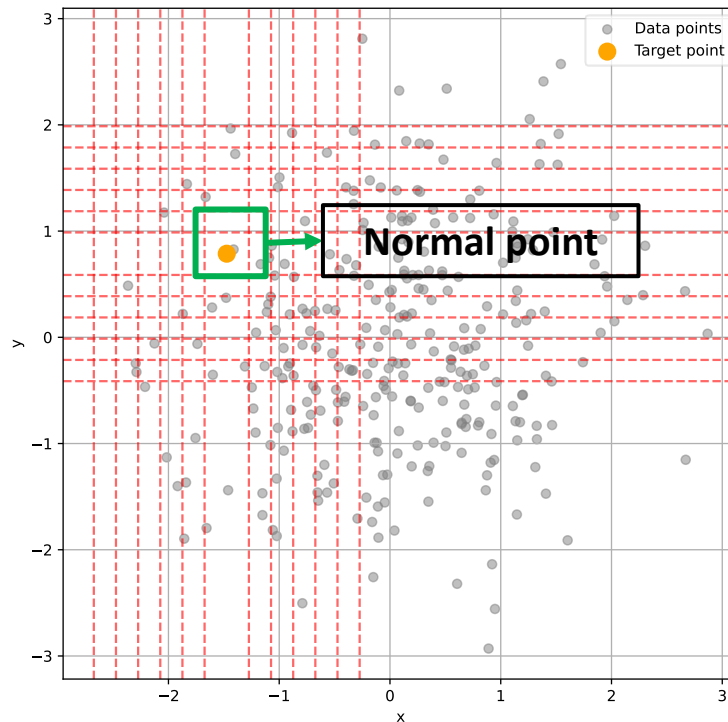
- 1) Random split** – choose a random feature and value to split data => build isolation trees
- 2) Building forest** – many trees with random data
- 3) Measuring isolation** – count how many splits are needed to isolate each point (path length)
- 4) Scoring:** shorter paths → higher anomaly score (0 to 1) → apply a threshold to detect anomalies.



- Anomalies require fewer splits to be isolated → **shorter path length**.

Isolation Forest for Anomaly Detection

Example of splitting a single point: more splits => normal point;
fewer splits => anomalous point



LSTM for Traffic Prediction

- LSTM unit architecture

At each step, the LSTM performs:

- **Forget gate** decides which parts of the old memory C_{t-1} should be removed using sigmoid.
- **Input gate** selects relevant new information via sigmoid and tanh.
- **Cell State Update**: combines filtered old memory and selected new values to form C_t .
- **Output gate**: filters C_t to produce hidden state h_t

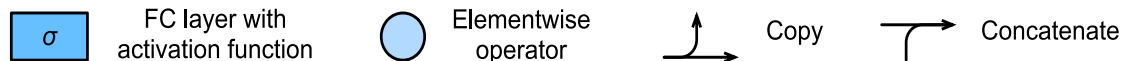
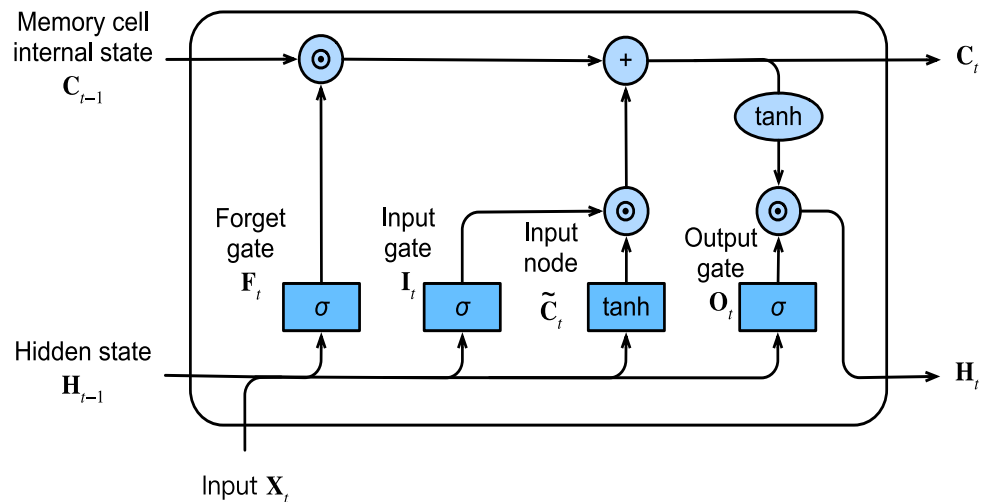


Figure 1. Architecture of an LSTM Unit
Adapted from Dive into Deep Learning [D2L23], licensed under Apache 2.0.
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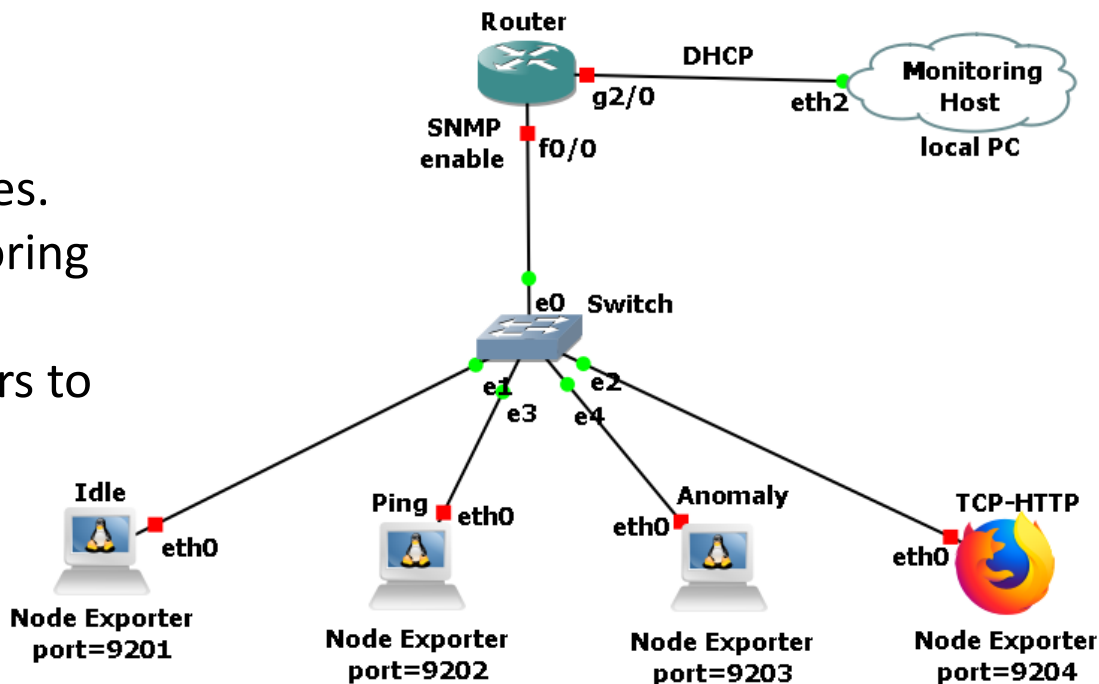
- Training using key SNMP network interface metrics (same as IF model)

Implementation

- GNS3 network architecture

Components:

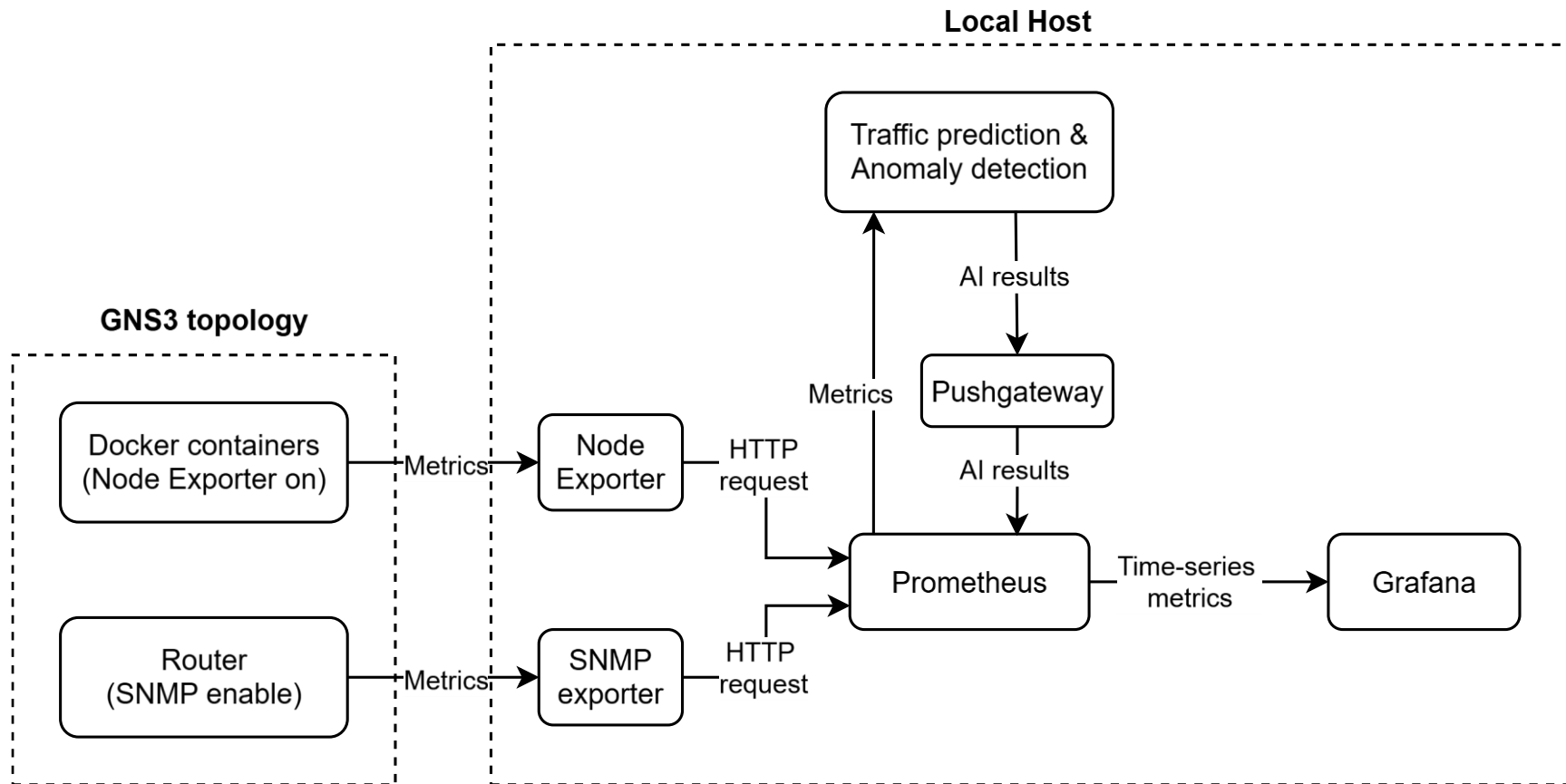
- Four Docker containers** to simulate different traffic types.
- Cisco Router** enables monitoring and acts as a gateway.
- Switch** connects all containers to the router.
- All containers run **Node Exporter** to provide system metrics.



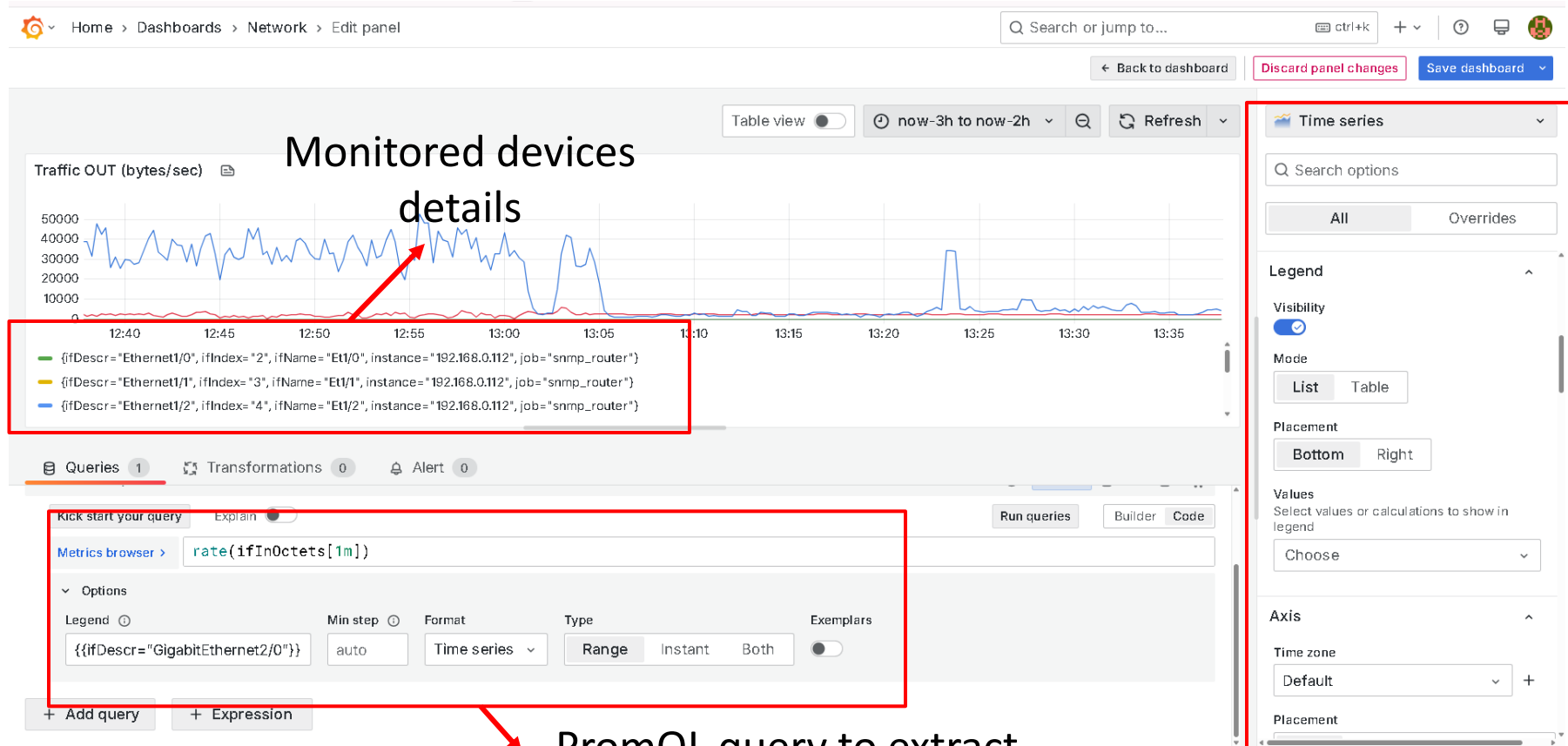
Router Configuration:

- SNMP enabled for interface metrics collection.
- Port forwarding (NAT) to access each container's Node Exporter port (e.g., 9201).
- Cloud node** (Monitoring Host) acts as a Bridged Mode between GNS3 topology and local PC providing Internet access

- Monitoring system architecture



- Grafana dashboard



PromQL query to extract metrics from Prometheus

Panel Editor

Implementation

- Prometheus configuration (snippet of yaml file):

```
global:
  scrape_interval: 15s

scrape_configs:
  - job_name: 'docker_nodes'
    static_configs:
      - targets:
        - '192.168.0.112:9201'
        - '192.168.0.112:9202'

  - job_name: 'pushgateway'
    static_configs:
      - targets: ['localhost:9091']

  - job_name: 'snmp_router'
    static_configs:
      - targets: ['192.168.0.112']
    metrics_path: /snmp
    params:
      module: [if_mib]
    relabel_configs:
      - source_labels: [__address__]
        target_label: __param_target
      - target_label: __address__
        replacement: localhost:9116
```

source label →

IP addresses to scrape metrics from →

source label →

GNS3 Router IP →

SNMP metrics path →

SNMP module (interfaces) →

SNMP Exporter local port →

Implementation

- SNMP metrics collection from GNS3 Router:
- OID** (Object Identifier) is a unique numeric identifier used in SNMP to locate and retrieve specific data from a device's MIB
- MIB** (Management Information Base) is a hierarchical database used in SNMP that maps OIDs to human-readable names

Metric name ← `- name: ifInOctets`
 Increases over time
 (cumulative count) ← `type: counter`
 Label used to identify the interface ← `labelname: ifIndex`
 Interface alias ← `labelname: ifAlias`
 Interface description ← `labelname: ifDescr`
 interface name ← `labelname: ifName`

```

oid: 1.3.6.1.2.1.2.2.1.10
help: The total number of octets
received on the interface,
including framing characters.
indexes:
- labelname: ifIndex
  type: gauge
lookups:
- labels:
  - ifIndex
- labelname: ifAlias
  oid: 1.3.6.1.2.1.31.1.1.1.18
  type: DisplayString
- labels:
  - ifIndex
- labelname: ifDescr
  oid: 1.3.6.1.2.1.2.2.1.2
  type: DisplayString
- labels:
  - ifIndex
- labelname: ifName
  oid: 1.3.6.1.2.1.31.1.1.1.1
  type: DisplayString
  
```


AI Integration in Network Monitoring System:

- AI scripts run every 60 seconds continuously via a **batch file** (run.bat)
- AI model input metrics:
 - ifHCInOctets – bytes received (shows how much data entered the router)
 - ifHCOctets – bytes sent (shows outgoing traffic from the router)
 - ifInErrors – errors on received packets (ex. connection issues)
 - ifOutErrors – errors on sent packets (ex. problems with the interface)
 - ifInDiscards – dropped incoming packets without errors (often due to overload)
 - ifOutDiscards – dropped outgoing packets (caused by congestion or limits)
- Input data are scaled (StandardScaler) before modeling:

```
scaler = StandardScaler()#scale to have mean=0 and std=1
```

- IF model was trained with contamination set to 0.1 (10%):

```
model = IsolationForest(contamination=0.1)#10% expected anomalies
```

- Traffic prediction (LSTM model) with **ReLU** activation function and sequence length of 10, trained with **MSE** loss and **Adam** optimizer:

```
model = tf.keras.models.Sequential([
    tf.keras.layers.LSTM(32,activation='relu',input_shape=(X.shape[1],
    X.shape[2])),tf.keras.layers.Dense(1) ])
model.compile(optimizer='adam', loss='mse')
```

Performance Metrics:

- Anomaly detection with IF:
 - `anomaly_rate` = percentage of data points detected as anomalies
 - `anomaly_score_mean` = average IF score (lower values = more unusual overall behavior)
 - `anomaly_score_max` = highest score in recent data (higher values = at least one point is normal)
- LSTM for traffic prediction:
 - Mean Squared Error (MSE) = Measures the average squared difference between predicted and actual values (lower = higher accuracy)
 - Predicted vs. real traffic graph (plotted in Grafana)
- Why not using accuracy?
 - No labeled data
 - Retraining on real-time data
 - MSE and anomaly scores were more relevant

Experimental Results

Scenario 1: GNS3 Topology without Monitoring and AI Facilities:

- Internet connectivity
- Traffic generated using:
 - ping commands:

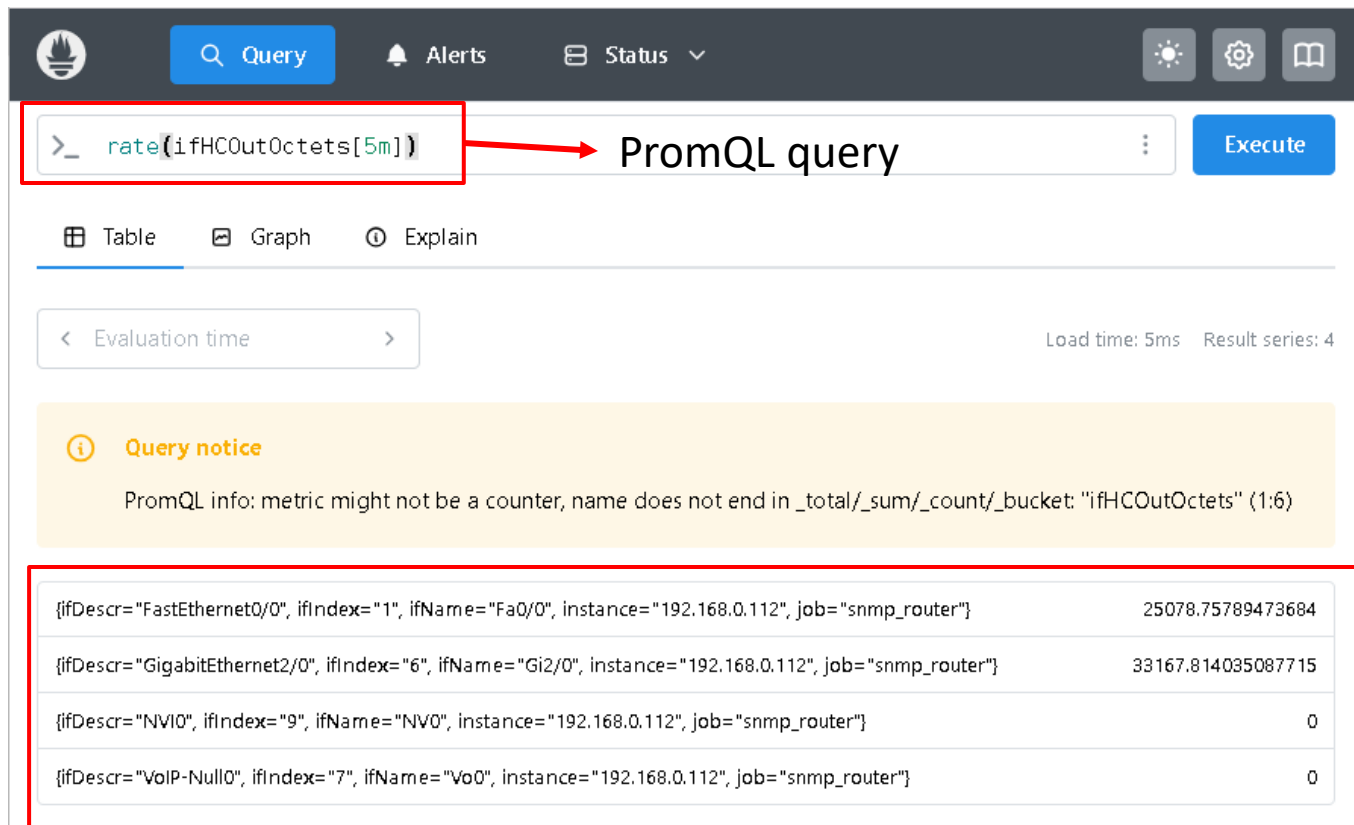
```
/ # ping -c 100 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: seq=0 ttl=117 time=13.0 ms
64 bytes from 8.8.8.8: seq=1 ttl=117 time=12.8 ms
```
 - webterm container for HTTP traffic type

Limitations:

- No monitoring or visibility
- No history (past data is lost)
- No alerts

Scenario 2: GNS3 Topology with Monitoring Capabilities

- Prometheus query displaying traffic for each router interface:



The screenshot shows the Prometheus web interface. At the top, there's a navigation bar with 'Query', 'Alerts', and 'Status' tabs. Below this, a query input field contains the PromQL query `>_ rate(ifHCOutOctets[5m])`, which is highlighted with a red box and labeled 'PromQL query' with a red arrow. To the right of the input field is an 'Execute' button. Below the input field, there are tabs for 'Table', 'Graph', and 'Explain', with 'Table' selected. A 'Query notice' box displays a warning: 'PromQL info: metric might not be a counter, name does not end in _total/_sum/_count/_bucket: "ifHCOutOctets" (1:6)'. Below the notice, a table displays the query results, which is also highlighted with a red box and labeled 'Query results displaying detailed interface traffic information' with a red arrow. The table has two columns: the first column contains JSON objects representing interface details, and the second column contains numerical values representing traffic rates.

{ifDescr="FastEthernet0/0", ifIndex="1", ifName="Fa0/0", instance="192.168.0.112", job="snmp_router"}	25078.75789473684
{ifDescr="GigabitEthernet2/0", ifIndex="6", ifName="Gi2/0", instance="192.168.0.112", job="snmp_router"}	33167.814035087715
{ifDescr="NV10", ifIndex="9", ifName="NV0", instance="192.168.0.112", job="snmp_router"}	0
{ifDescr="VoIP-Null0", ifIndex="7", ifName="Vo0", instance="192.168.0.112", job="snmp_router"}	0

Experimental Results

- Network metrics provided by Prometheus visualized in Grafana



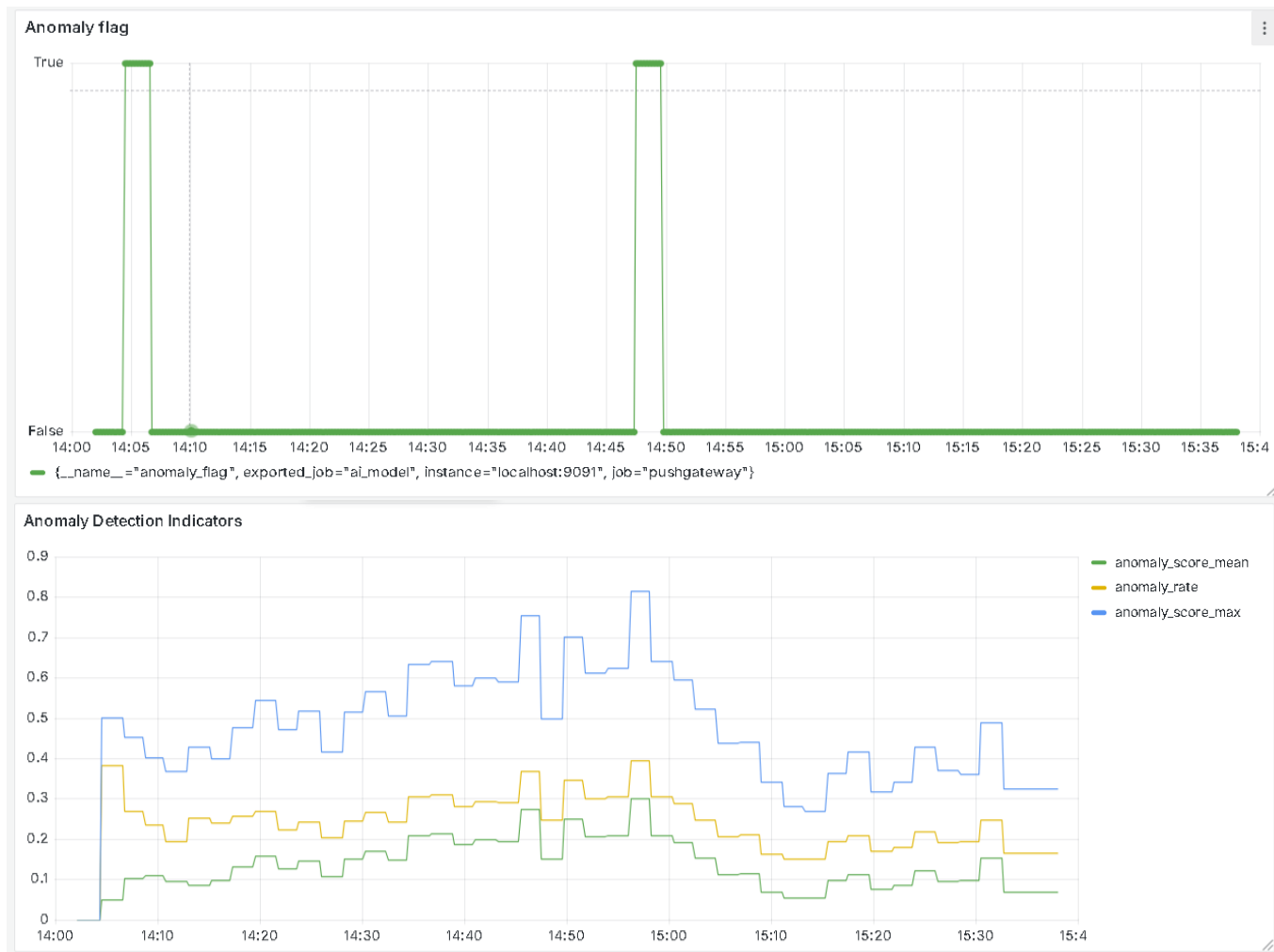
Scenario 3: Network Topology with Monitoring and AI Integration

- Predicted traffic vs Real traffic and MSE visualized in Grafana



Experimental Results

- Anomaly detection and Indicators visualized in Grafana



Conclusions and Future Developments

- **Conclusions**

- Monitors and analyzes network traffic in real-time (with/without Internet connection)
- Uses AI for anomaly detection and traffic prediction
- Built with open-source tools, easy to integrate
- Flexible and free alternative to commercial monitoring tools

- **Future Improvements**

- Train LSTM model with more data for better predictions
- Test other AI models like GRU, CNN, and Random Forest
- Reduce false alerts by tuning anomaly detection settings
- Add more devices and traffic types (DNS, file transfers, streaming) in the network setup for more realistic testing

- [Alb22] L. Alberro, A. Castro, E. Grampin, "Experimentation Environments for Data Center Routing Protocols: A Comprehensive Review", *Future Internet*. 2022; 14(1):29. <https://doi.org/10.3390/fi14010029>
- [Alt24] E. Altulahan, M. A. Almaiah și A. Aljughaiman, "Anomaly Detection IDS for Detecting DoS Attacks in IoT Networks Based on Machine Learning Algorithms," *Sensors*, vol. 24, art. nr. 713, 2024, doi: 10.3390/s24020713
- [Cal22] O. Calzone, "An Intuitive Explanation of LSTM," *Medium*, Feb. 21, 2022. [Online]. Available: <https://medium.com/@ottavioalzone/an-intuitive-explanation-of-lstm-a035eb6ab42c>
- [Cir25] S.D. Cimat, V. Dobrota, "Isolation Forest Anomaly Detection and Long Short-Term Memory Traffic Prediction", 20th Student Symposium on Electronics and Telecommunications SSET 2025, Technical University of Cluj-Napoca, 2025.
- [Emm23] I. D. Emmanuel, N. Linge and S. Hill, "Analysis of SD-WAN Packets using Machine Learning Algorithm," 2023 Conference on Information Communications Technology and Society (ICTAS), Durban, South Africa, 2023, pp. 1-6, doi: 10.1109/ICTAS56421.2023.10082743.
- [Fon18] M. Fomin, "GNS3 for Network Emulation", *International Journal of Engineering and Technology*, vol. 7, no. 4, pp. 317–320, 2018.
- [GNS25] "GNS3 — Graphical Network Simulator," GNS3.com, 2025, [Online]. Available: <https://www.gns3.com>
- [Gra25] "Introduction to Grafana," Grafana Documentation, 2025, [Online]. Available: <https://grafana.com/docs/grafana/latest/>
- [Gre17] D. Greff, R. K. Srivastava, J. Koutník, B. R. Steunebrink, and J. Schmidhuber, "LSTM: A Search Space Odyssey," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 28, no. 10, pp. 2222–2232, 2017, doi: 10.1109/TNNLS.2016.2582924.
- [Jai24] M. Jain și A. Shah, "Anomaly Detection Using Convolutional Neural Networks," *ESP International Journal of Advancements in Computational Technology*, vol. 2, nr. 3, pp. 12–22, 2024, doi: 10.56472/25838628/IJACT-V2I3P102
- [Lal20] B. Lal, A. Patel, R. Sharma and R. Thakur, "Survey of Anomaly Detection Techniques in Data Mining", *International Journal of Innovative Technology and Exploring Engineering (IJITEE)*, vol. 9, no. 1, pp. 3799–3803, Nov. 2019.
- [Liu23] Y. Liu, S. Zhang, and L. Song, "Learning to Improve Isolation Forest," *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence (IJCAI-23)*, held in Macao, S.A.R., 2023, pp. 2383–2390. International Joint Conferences on Artificial Intelligence Organization. [Online]. Available: <https://www.ijcai.org/proceedings/2023/0264.pdf>
- [Lyp24] A. Lypka, D. Malowany, and J. Domanska, "Benchmarking Feature Extraction Tools for Network Traffic Classification," *Scientific Reports*, vol. 14, no. 1, Apr. 2024, Article no. 70983. doi: 10.1038/s41598-024-70983-6.
- [OID25] "Simple Network Management Protocol: OID Overview and MIBs", Global OID Reference Database, 2025, [Online]. Available: <https://oidref.com>
- [Pou24a] N. Poulton, "Docker Deep Dive. Zero to Docker in a single book!", Nigel Poulton and Lean Publishing, 2024.
- [Pou24b] N. Poulton, "The Kubernetes Book", Nigel Poulton and Lean Publishing, 2024.
- [Pro25] "Prometheus: Monitoring system and time series database," Prometheus.io, [Online]. Available: <https://prometheus.io/>
- [Pyn25] "Top 10 Most Popular Network Simulation Tools (2025)," PyNet Labs, 2025, [Online]. Available: <https://www.pynetlabs.com/top-most-popular-network-simulation-tools/>
- [Sal24] A. A. Salahuddin, M. F. Bari, I. A. Qazi, and R. Boutaba, "Machine Learning for Classification of SDN Traffic," *arXiv preprint, arXiv:2501.13004*, Jan. 2024. [Online]. Available: <https://arxiv.org/abs/2501.13004>
- [Tur19] Y. Turk, E. Zeydan and Z. Bilgin, "A Machine Learning Based Management System for Network Services," 2019 International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob), Barcelona, Spain, 2019, pp. 1-8, doi: 10.1109/WiMOB.2019.8923572.
- [Tur24] F. Turkal, "How does Docker ACTUALLY work? The Hard Way: A Comprehensive Technical Deep Diving," *Medium*, Jan. 4, 2024. [Online]. Available: <https://medium.com/@furkan.turkal/how-does-docker-actually-work-the-hard-way-a-technical-deep-diving-c5b8ea2fd422>
- [Wen21] Y. Wen, Y. Liu, C. Lin and J. Pan, "Point-Denoise: Unsupervised Outlier Detection for 3D Point Clouds Enhancement," *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 14660–14669, doi: 10.1109/CVPR46437.2021.01444.
- [Wik25a] "Long short-term memory," Wikipedia, The Free Encyclopedia, 2025, [Online]. Available: https://en.wikipedia.org/wiki/Long_short-term_memory
- [Wik25b] "Isolation Forest," Wikipedia, The Free Encyclopedia, 2025, [Online]. Available: https://en.wikipedia.org/wiki/Isolation_forest
- [Xia18] C.-H. Xiao, C. Su, C.-X. Bao and X. Li, "Anomaly Detection in Network Management System Based on Isolation Forest," 2018 4th Annual International Conference on Network and Information Systems for Computers (ICNISC), Zhangjiajie, China, 2018, pp. 56–60, doi: 10.1109/ICNISC.2018.00019.
- [Zha22] J. Zhao, X. He, "NTAM-LSTM Models of Network Traffic Prediction," *MATEC Web of Conferences*, vol. 355, art. nr. 02007, 2022, doi: 10.1051/mateconf/202235502007.
- [Zha23] A. Zhang, Z. C. Lipton, M. Li, and A. J. Smola, *Dive into Deep Learning*, Cambridge University Press, 2023, [Online]. Available: <https://d2l.ai>

Questions?



